PH.45320 **FALL 2026**

Static Cosmology

Lecture 9: Distance-Dependent Extinction

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Overview

A smooth distance-dependent opacity component is estimated without absorbing source-population trends.

1 Observational setup

Source coordinates, observing band, selection function, and calibration state are fixed at the outset.

$$\kappa_\nu(s) = \kappa_{0,\nu} + \kappa_{1,\nu} G(s; s_*, w)$$

2 Transport or equilibrium equation

The governing equation is solved first in a homogeneous limit and then with the stated spatial dependence.

$$F_{obs} = F_0 e^{-\int \kappa_\nu(s)\rho_m ds}$$

3 Connection to data

Each model quantity is tied to a measured flux, spectrum, angle, count, or temperature.

$$G = (1 + e^{-(s-s_*)/w})^{-1}$$

4 Degeneracies

Calibration, population, and medium parameters are varied separately to expose their degeneracies.

5 Example

The worked example carries units and uncertainty from the input catalog to the reported result.

Checks

- Verify dimensions and sign conventions.
- State the boundary conditions used in each limit.
- Report numerical precision separately from model uncertainty.

Reading

- X-7 release
- Jeong, ch. 9
- CMT extension protocol